CLOAKD

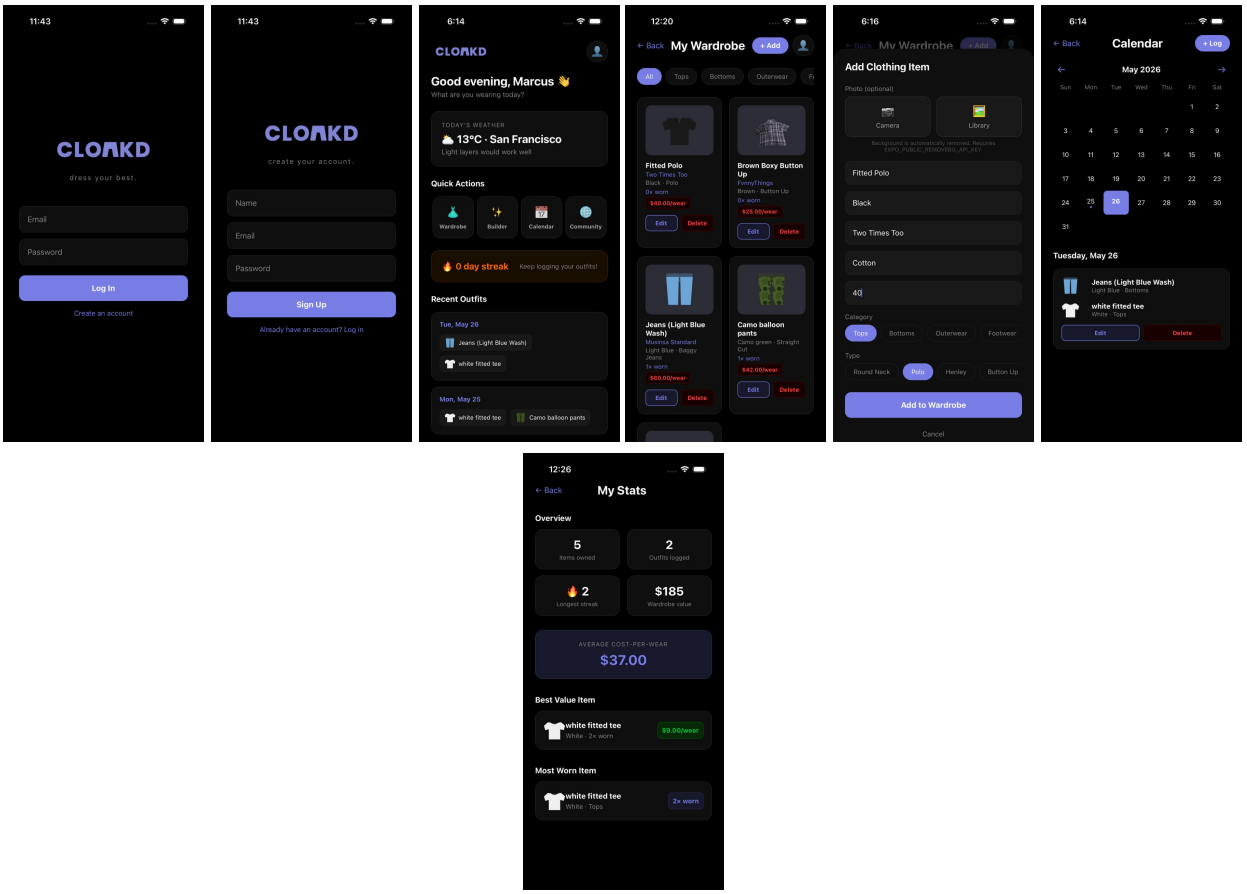
Dress your best.

Milestone 1 Submission

Team Name: CLOAKD

Proposed Level of Achievement: Apollo 11

Proof of Concept



Motivation

For most people, us included, getting dressed is stressful. You stand in front of a full wardrobe and somehow still feel like you have nothing to wear. CLOAKD reframes that daily moment as something expressive and enjoyable instead of a chore and removes the anxiety from picking out an outfit.

Most people wear fewer than a third of the clothes they own. The average piece of clothing gets worn just 7 times before it's thrown away and 40% of what we buy never gets worn at all. People keep buying new clothes and feeling like they constantly have nothing to wear not because they lack options, but because they can't see what they already have. CLOAKD solves a **visibility problem, not a shopping problem.**

Fast fashion is one of the most polluting industries in the world, but telling people to "buy less" doesn't work. CLOAKD takes a different approach by making it fun, social and rewarding to get more out of the clothes you already own. The cost-per-wear tracker and wardrobe utilisation score turn sustainability into a game rather than a guilt trip.

We also wanted to build something that brings people together around style in a way that feels real and also to learn how to take an idea from nothing and turn it into something real while picking up software engineering skills along the way by building something we actually care about.

Vision

CLOAKD will be a mobile app for iOS and Android, built with React Native and Expo. It lets users digitise their entire wardrobe and receive daily outfit suggestions built entirely from clothes they already own, filtered by real-time weather and occasion.

The app is designed around a simple daily loop: open the app, get a suggestion tailored to today's weather and occasion, log what you wore and watch your wardrobe come alive through stats, streaks and community sharing.

Ultimately, we envision CLOAKD as the app people reach for every morning, the one that changes how they relate to the clothes they already own, reduces mindless overconsumption and makes getting dressed something enjoyable.

User Stories

1. As a person who wants to dress better and improve my style, I want to be able to visualise and plan my outfits.
2. As a person who struggles with deciding what to wear, I want to be able to receive outfit suggestions for the day.
3. I want to be able to dress well and look stylish no matter the weather, even if it is a blazing hot day or a rainy day with thunderstorms. I also want to be able to dress appropriately for the occasion, such as date nights, formal occasions or gym days.
4. I want to be able to take inspiration from others' outfits and share my own.
5. As a frugal person, I want to be able to track how much value and how many wears I am getting out of my clothes.
6. I want to keep track of the outfits I have worn each day so that I do not repeat outfits too frequently.
7. As someone with a large wardrobe, I want to be able to search and filter my clothes by colour, category or material so that I can quickly find what I'm looking for without digging through my closet.
8. As someone who hasn't worn certain pieces in a long time, I want to be reminded of clothes I've forgotten about so that I can rediscover and make better use of them.
9. As someone who is trying to be more intentional about shopping, I want to see which gaps actually exist in my wardrobe before buying anything new.

10. As a conscious consumer, I want to understand the true cost of my purchases over time so that I can justify spending more on quality pieces that I'll wear repeatedly.
11. As someone who enjoys fashion, I want to browse trending outfits in the community and save looks that inspire me so that I have a personalised mood board to reference when planning my own outfits.
12. As someone who enjoys tracking personal progress, I want to look back at my outfit history over weeks and months so that I can see how my personal style has evolved.

Scope of Project

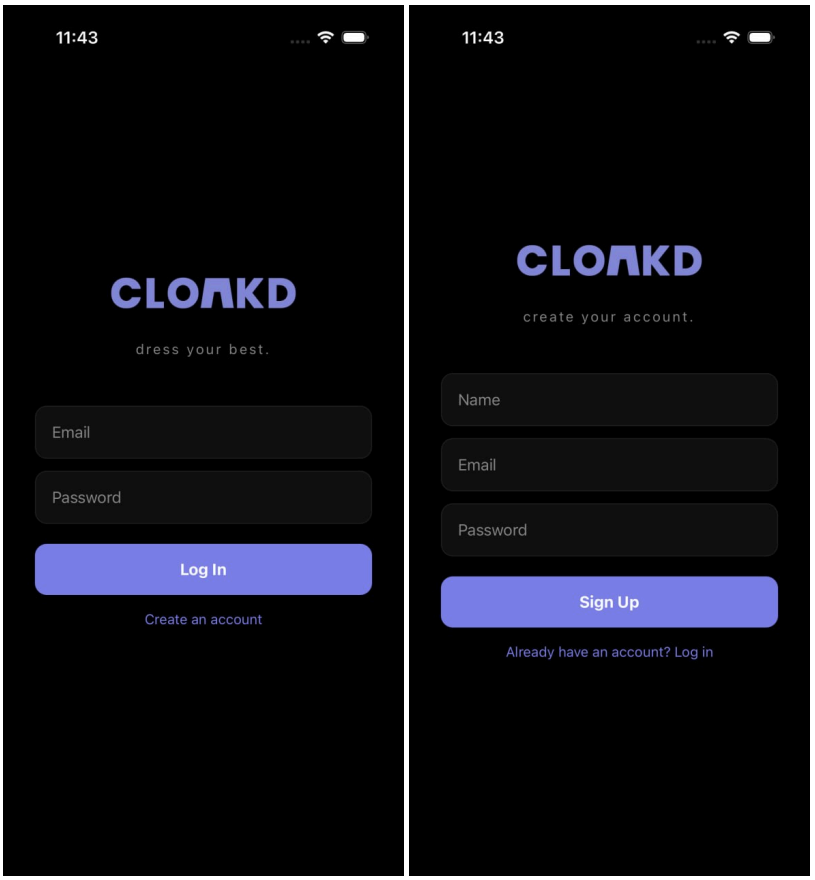
CLOAKD is a mobile app that lets users photograph and catalogue every clothing item they own, then suggests daily outfits built entirely from their existing wardrobe — filtered by real-time weather and occasion. Users log outfits to a calendar, track cost-per-wear statistics, build daily streaks, and optionally share looks to a community feed.

CLOAKD's features are outlined in the following sections, organised by the following tags:

- **[Proposed]** — features targeted for completion
- **[Current Progress]** — what has been built so far

Features

User Account Authentication



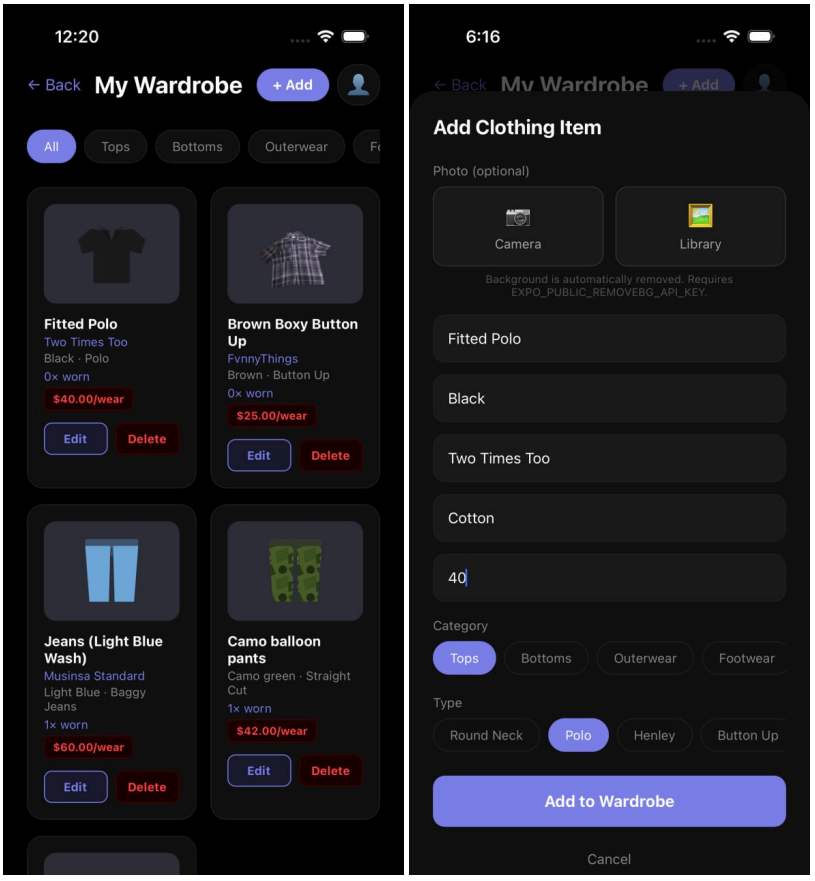
[Proposed]

Each user will be authenticated with their own unique account. Utilising the Firebase SDK Authentication, users can log in via Email and Password. They will be able to create an account if the user does not already have an existing account. Based on the user's action, the relevant call to the Firebase authentication instance is made. The FirebaseAuthExceptions are handled within a try-catch block and an alert bar is displayed when the user keys in an invalid input. Only upon successful login or account registration will the user be routed to the next page and the account reflected in the Firebase authentication.

[Current Progress]

Users are able to either log in with their existing account if they have already made one or sign up for a new account with their name, email and password. Upon inputting their log in information, if their account exists and their information is correct, they will be redirected to the home page of their account.

Virtual Wardrobe



[Proposed]

Users add their clothes by uploading a photo of each item and tagging it with basic details such as colour, category, subcategory, material, brand and purchase price. The wardrobe screen displays all items at a glance, filterable by category, colour or occasion in one tap. Items that have not been worn for 30 or more days and new items that have been added but are yet to be worn are surfaced automatically to encourage rediscovery.

[Current Progress]

Users can save clothing items with basic tags (category, subcategory, colour, material, brand and purchase price) to the database via Supabase. The wardrobe grid view is functional and displays all saved items. Image

capturing and background removal has been implemented. Filtering and the 30-day rediscovery have been implemented as well.

Outfit Builder

[Proposed]

The outfit builder suggests outfit combinations built entirely from clothes the user already owns. The user is presented with a top, bottom, outerwear, footwear and accessories slot. Each slot is independently swappable, meaning the user can keep what they like and shuffle individual pieces. Weather and occasion filters are applied before suggestions load. Finalised outfits can be saved and logged.

[Current Progress]

The outfit builder feature has not been made yet.

Occasion and Weather Filter

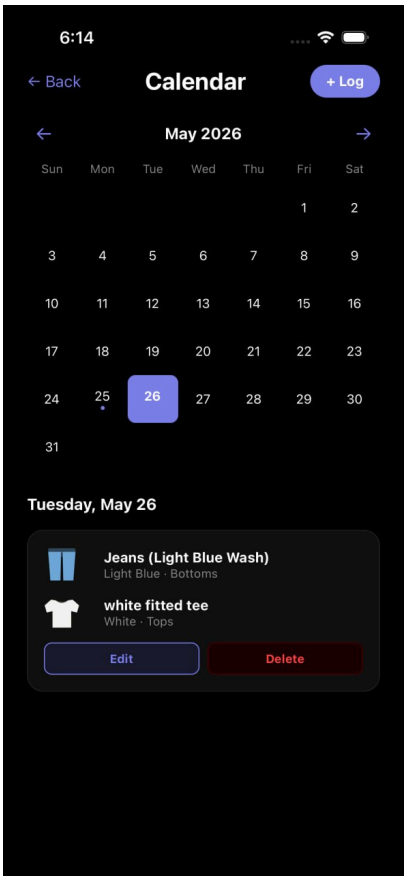
[Proposed]

The filter asks the user where they are going and what they will be doing. It uses real-time weather data from the Open-Meteo API for the user's location and filters the wardrobe to suggest outfits suitable for the conditions and occasion. Supported occasions: casual, office, formal, outdoor, date night.

[Current Progress]

Open-Meteo API integration is complete and returning live weather data for the user's current location. The logic to filter wardrobe items by temperature range and occasion tag has not yet been connected to the outfit builder.

Outfit Calendar



[Proposed]

The outfit calendar acts as a personal style diary. Every time a user logs an outfit, it is recorded against the specified date. Users can look back to see exactly what they wore and when. Logging an outfit automatically updates the wear count and cost-per-wear for every item in the outfit. The calendar also surfaces pieces that have not been worn in a while to encourage rediscovery. Users can also edit and delete outfits that have been logged previously.

[Current Progress]

The calendar has been created and can be navigated by month. Logging an outfit to a date is functional and stored in the database. Automatic updating of wear counts on log is complete. Visual indicators for logged days as well as current date are displayed on the calendar grid. Clothing items that have not been worn in a while have not been set to be suggested yet.

Share Outfits

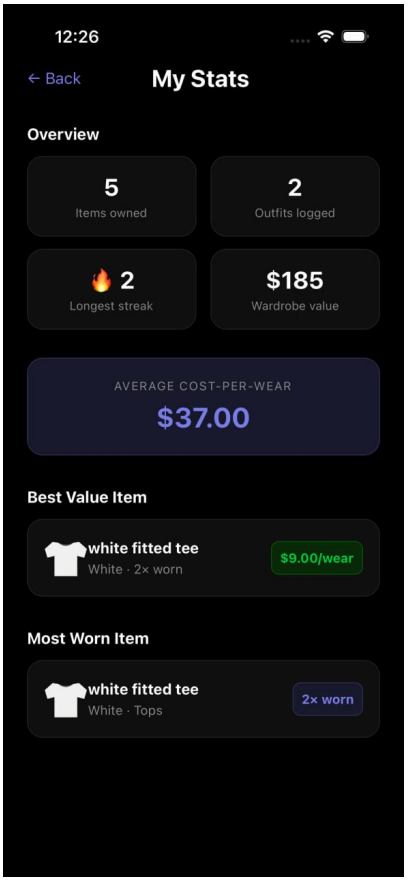
[Proposed]

Users can opt in to post outfits to the CLOAKD community feed. Privacy is the default, meaining that a user's wardrobe stays invisible unless they choose to share the outfits they build from it. Shared posts display only garment categories and colours, never purchase prices or private tags. Other users can like and save posts to their inspiration board.

[Current Progress]

The community feed has not been built yet.

Cost-Per-Wear Tracker



[Proposed]

When adding clothing items, users input the purchase price. Cost-per-wear is calculated as price divided by the number of wears and updated automatically each time the item is logged as part of an outfit. The wardrobe view surfaces this figure alongside wear count for each item, giving users a clear picture of the value they are getting out of their clothes.

[Current Progress]

The cost-per-wear tracker has been fully built. Users can easily see the number of times they worn each piece of clothing and the cost-per-wear of it in the wardrobe page. Adding a clothing item to an outfit and logging it in the calendar automatically updated the wear count and cost-per-wear of that specific clothing item. There is also a stats page which shows users their best value clothing item (the one with the lowest cost-per-wear) and other statistics like their average cost-per-wear and most worn item.

Streak System

[Proposed]

Every day a user logs an outfit, their streak grows. Missing a day resets the counter. Badges unlock at 10-day, 50-day and 100-day milestones to keep users motivated to open the app daily.

[Current Progress]

A basic streak counter has been implemented when a user logs an outfit in the calendar. The streak counter on the home page is yet to be updated. The badges have not been made nor has the badges page been built.

Trending Outfits

[Proposed]

A weekly feed surfaces the most popular community outfits, ranked by an algorithm combining recency, saves and wardrobe diversity. This gives users fresh inspiration and rewards creativity with existing clothes over purchasing new ones.

[Current Progress]

The trending outfits algorithm has not been built yet.

Timeline and Development Plan

Milestone	Date	Task & Description
Liftoff	10 - 18 May 2026	Create project poster.
Liftoff	10 - 18 May 2026	Film, edit and craft liftoff project poster.
Liftoff	10 - 18 May 2026	Finalise project features and review proposal.

> **Evaluation Liftoff: 18 May 2026**
>
> - Create project poster and video.
> - Solidify plans for CLOAKD app.
> - Self-learn all the tools in our tech stack to build our CLOAKD app.

Milestone	Date	Task & Description
1	19 - 24 May 2026	Set up Expo project, Firebase backend and Supabase backend. Integrate Expo with Firebase for user authentication and Supabase for clothing item information.
1	19 - 24 May 2026	Create log-in and sign-up pages with user email and password for authentication and verification.
1	19 - 24 May 2026	Create home page as a dashboard that is connected to all other pages and features.
1	24 - 31 May 2026	Create virtual wardrobe feature with image capture and input of clothing item details such as colour, material, subcategory and purchanse price. Link to Supabase for storage of details.
1	24 - 31 May 2026	Create outfit calendar feature with ability to log outfits and highlight dates where outfits have been logged.
1	24 - 31 May 2026	Create Cost-Per-Wear extension to update wear count every time the clothing item is logged in an outfit. Use purchase price / number of wears to calculate.

> **Evaluation Milestone 1: 1 June 2026**

> - Able to add clothes to the virtual wardrobe + store and access data.
> - Outfit Calendar to be functional.
> - Calculation feature of Cost-per-wear Tracker to be functional. Integration with Virtual Wardrobe to be completed at a later date.
> - Account sign-up and log-in with authentication to be completed.
> - Basic UI complete.

Milestone	Date	Task & Description
2	2 - 15 June 2026	Create outfit builder feature and link it to the virtual wardrobe to directly access all clothing items.
2	2 - 15 June 2026	Add function that pushes clothes that have not been worn in a while for outfit builder feature.
2	2 - 15 June 2026	Include occasion and weather filter to suggest outfit recommendations for outfit builder feature.
2	16 - 28 June 2026	Create settings page.
2	16 - 28 June 2026	Start creating community feed and outfit sharing feature.

> **Evaluation Milestone 2: 29 June 2026**
>
> - Occasion and Weather Filter to be functional.
> - Outfit builder to be functional.
> - Backend of Virtual Wardrobe completed. Integration with Cost-per-wear Tracker.
> - Frontend of Share Outfits done.

Milestone	Date	Task & Description
3	1 - 26 July 2026	Complete building streak system.
3	1 - 26 July 2026	Create and add achievement badges for hitting certain levels of streaks, outfits logged, achieving certain levels of value for Cost-Per-Wear and number of clothing pieces added.
3	1 - 26 July 2026	Ensure that community feed feature is fully functional.
3	1 - 26 July 2026	Implement trending outfit feature in feed and include trending outfit calculator algorithm based on views, likes, saves and shares.
3	1 - 26 July 2026	Create outfit sharing feature and ensure it is fully functional.

> **Evaluation Milestone 3: 27 July 2026**
>
> - Streak System to be functional + badges/achievement tokens to be designed.
> - Share Outfits feature to be functional.

> - Trending Outfits feature to be functional.
> - Community feed feature to be fully functional.

Milestone	Date	Task & Description
Splashdown	1 - 25 August 2026	Fix all bugs and improve UI.
Splashdown	1 - 25 August 2026	Make quality of life updates and improvements.

> **Evaluation Splashdown: 26 August 2026**
>
> - Fix all bugs and improve UI.
> - Quality of life updates and improvements.

Database Schema



Tech Stack

Layer	Technology used
Frontend Design & App Framework	React Native, Expo
Backend Design	Node.js, Express.js
PostgreSQL Database	Supabase
Auth & Push	Firebase
Caching	Upstash Redis
Weather API	Open-Meteo
Version Control	Git, Github

Project Log

Project Log

Project Poster

Project Poster

Project Video

Project Video

Dress your best. · *Team CLOAKD*

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